Rijaul Haque

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About Me

I am a **B.Tech** candidate in **Computer Science and Engineering** at Dibrugarh University, with a **Minor in Artificial Intelligence and Machine Learning**. My primary interests lie in the mathematical and theoretical foundations of Artificial Intelligence—particularly across machine learning, deep learning, and reinforcement learning. Driven by a curiosity to understand and optimize the mechanics behind these models, I am actively seeking hands-on research environments where I can collaborate with scholars, explore emerging paradigms, and contribute to meaningful advancements in theoretical AI.

Education

Dibrugarh University *Aug'22 – Aug'26*
B.Tech in Computer Science and Engineering

- CGPA: 8.482/10.0 (After 7th Semester)
- Minor specialization in Artificial Intelligence and Machine Learning
- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Computer Organization & Architecture, Operating Systems, Principles of Programming Language, Theory of Computation, Compiler Design, AI(Knowledge & Representation), Machine Learning, Deep Learning, Practical Reinforcement Learning, Cryptography & Network Security, Software Engineering.

Research Interests

- **Theoretical Machine & Deep Learning:** Exploring the foundational mechanics, mathematical optimization, and architectural design of ML/DL models to build more robust, generalizable AI systems.
- **Reinforcement Learning & Decision Systems:** Investigating exploration-exploitation dynamics, specifically within Contextual Multi-Armed Bandits, to advance scalable and sample-efficient sequential decision-making.
- **Multimodal AI & Sensor Fusion:** Researching architectures capable of integrating diverse data modalities (such as 3D spatial data, vision, and text) to enhance holistic scene understanding and predictive modeling.

Experience

Web Development & Networking Intern *Assam, India*
BCPL (A Govt. of India Enterprise), India *Jan 2025 (1 mo.)*

- Engineered backend logic and data interfaces using the ASP.NET framework, focusing on system reliability and efficient data routing.
- Analyzed enterprise-level network management protocols and topology, building a practical foundation in network optimization and distributed environments.

Blockchain Technology Intern *Meghalaya, India*
NIT Meghalaya, India *Jun 2024 - Jul 2024 (2 mos.)*

- Architected a decentralized booking system using Blockchain technology, focusing on secure state management, cryptographic protocols, and distributed data integrity.
- Gained hands-on experience with consensus algorithms and distributed systems architecture, providing a strong conceptual groundwork for exploring different aspects of decentralized system.

Projects

LiDAR & RGB-D Multimodal Sensor Fusion System (Ongoing) *Sep 2025 – Present*

- Engineered a hands-on multimodal data pipeline, successfully achieving high-fidelity **3D color-depth fu-**

sion by spatially aligning sparse LiDAR point clouds with RGB-D visual streams.

- Implemented complex point cloud registration and mathematical projection algorithms to accurately have dense 3D spatial geometry.
- Currently utilizing this synthesized 3D environment to research, train, and evaluate **deep learning architectures** for advanced spatial reasoning and physical scene understanding.
- Tools/Tech: **Python, Open3D, OpenCV, Point Cloud Library (PCL), Robot Operating System(ROS2), Ubuntu, 3D Computer Vision, Deep Learning**.

Comparative Analysis of Contextual Bandits for Movie Recommendation *Sep 2025 – Dec 2025*

- Engineered and evaluated a real-time recommendation engine using **Contextual Multi-Armed Bandits (CMAB)** to solve the "Cold Start" problem.
- Implemented and benchmarked three distinct RL strategies—**LinUCB**, **Thompson Sampling**, and a **novel Stochastic Reward formulation**—on the MovieLens 100k dataset.
- Developed a **Stochastic Reward environment** to model real-world user passivity, maximizing engagement volume with a **53.42% CTR**.
- Conducted a comparative performance analysis using **Precision@5** and **Cumulative Regret** metrics, establishing a theoretical performance ceiling via **Singular Value Decomposition (SVD)**.
- Tools/Tech: **Python, NumPy, Pandas, Scikit-learn (SVD), Reinforcement Learning (CMAB), Matplotlib.**

AI-Powered Career Analytics & LLM Integration *Dec 2025*

- Integrated large language models (**DeepSeek-v3.1 via OpenRouter API**) to dynamically process user profiles and synthesize structured analytical output.
- Architected a robust data persistence layer using **Spring Data JPA** and **MySQL** to handle complex, long-term user analytics and state management.
- Tools/Tech: **Java, LLM APIs, Spring Boot, MySQL, REST Architecture**.

Relational Database Optimization for Academic Systems *Jan 2025*

- Designed and optimized relational database schemas for a high-throughput attendance tracking system, focusing on efficient data retrieval and secure authorization roles using **SQL** and **ASP.NET**.

Skills

Programming Languages: Java, Python, C, C++, Go, JavaScript, HTML/CSS.

Frameworks & Libraries: Spring Boot, Spring Data JPA, WebFlux, TensorFlow, ASP.NET, Pandas, NumPy, NLTK, Scikit-Learn, Matplotlib, Open3D, OpenCV

Tools & Technologies: Git, Docker, AWS, MySQL, Linux, Hyperledger Fabric, OpenRouter API, iText (PDF), ROS2 jazzy.

Web/Frontend: Thymeleaf, REST APIs, HTML/CSS/JS

Soft Skills: Problem Solving, Team Collaboration, Project Management, Technical Documentation

Certifications

Mathematical Foundations for Machine Learning [View] *Oct 2025*
NPTEL (Indian Institute of Science Bangalore)

Programming with Generative AI [View] *Oct 2025*
NPTEL (Indian Institute of Science Bangalore)

Machine Learning Specialization [View] *Oct 2024*
Coursera (Stanford University)

Linear Algebra [View] *Mar 2026*
Coursera (Johns Hopkins University)

Achievements & Volunteering Contributions

- Ranked in the **Top 5%** of candidates in the NPTEL course "Programming with Generative AI" with a consolidated score of 80%, certified by IISc Bangalore.
- Volunteered at the National Conference on Multidisciplinary Approaches in Networks, and Computing (NCMNC-2025) organized by Dibrugarh University.
- Active IEEE student member.
- Qualified and participated in the State-level Science-based Extempore Speech (Native Language) Competition organized by the Assam Science Technology & Environment Council, Govt. of Assam.